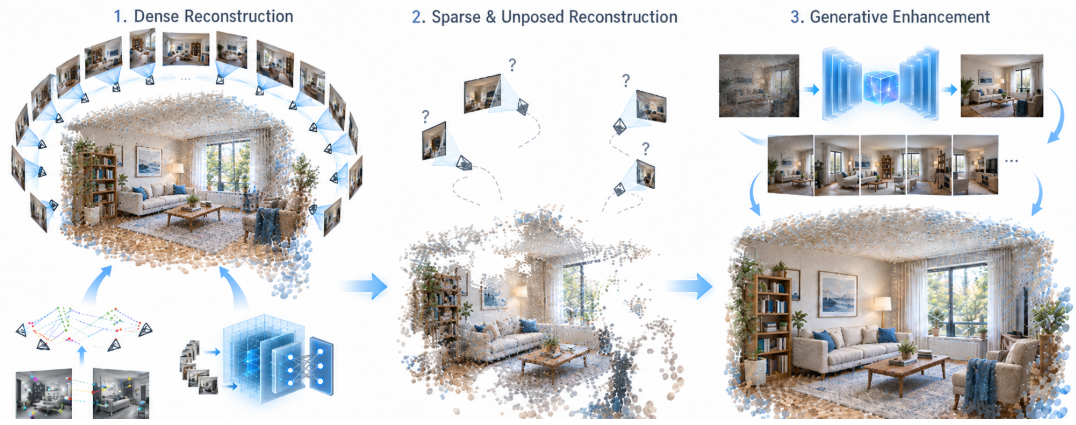
Project 4: Generative Sparse-View 3D Reconstruction

AIAA 3201—Introduction to Computer Vision

Term Project Instruction

Fall 2026



1 Overview

Core Task: The goal of this project is to experience the evolution of 3D reconstruction from "ideal" to "extreme" conditions, and to understand how modern AI priors are changing the reconstruction pipeline.

1. Start with **ideal conditions** (dense data) to compare classical multi-view geometry with modern 3D Foundation Models, and study how data-driven scaling affects reconstruction.
2. Challenge **extreme conditions** (unknown poses, sparse views) using unposed reconstruction or monocular 3DGS-SLAM without ground truth poses.
3. Explore **Generative AI** to restore or generate missing visual information, thereby improving sparse-view reconstruction.

Key Concepts:

- **3D Reconstruction:** 3D Gaussian Splatting (3DGS).
- **3D Foundation Models:** Classical SfM (COLMAP) vs. feed-forward geometry models (VGGT/VGGT-Ω).
- **Unposed Reconstruction:** Pose estimation, global consistency, optimization, and monocular 3DGS-SLAM.
- **Generative AI:** Image and video diffusion priors for sparse-view reconstruction.

Group Size: 2 students per group.

2 Submission Requirements

- **PDF Report:**
  - **Format:** Use the official CVPR LaTeX Template. The length must be **6–8 pages** (excluding references).
  - **ArXiv Upload (Optional):** You may upload your final report to **arXiv** if desired.
  - **Content Requirements:**
    1. **Abstract & Introduction:** Background, motivation and solution overview. **Explicitly include your GitHub repository link at the end of abstract.**
    2. **Related Work:** Review and cite the core papers listed in the Recommended Reading List (Section 4).
    3. **Method:** Explain the technical roadmaps **in your own words**.
    4. **Experiments:** Present quantitative results (Tables) and qualitative analysis (Figures).
    5. **Conclusion:** Summary, limitations, and future work.
- **Code (Mandatory):** Upload your code to a public **GitHub repository** with a clear `README.md` explaining how to run it. Do not submit raw code files. Include dependencies, usage, weights, and visual results in your README.
- **Demo Video (Bonus):** Pack your rendered video results into a `.zip` file and upload it to Canvas. Pack them into a zip file named `videos.zip` and upload it to Canvas. High-quality visualizations will receive **bonus points**.

Important Tips for Success

- **Method Flexibility:** You are encouraged to follow our suggested roadmap, but you are also free to design your own optimal method. Strict adherence to the "Implementation Roadmap" is not mandatory as long as the core tasks are fulfilled.
- **Completeness:** All three parts are **mandatory**. If you fail to complete Part 3, you must discuss the reasons and demonstrate failure cases to receive decent score. Ignoring it completely will result in an "Incomplete" grade.
- **Visuals:** Clean and aesthetic visualizations (e.g., flowcharts, rendering comparisons) are crucial for a Computer Vision course and will significantly improve your score.
- **Dataset Scope:** Completing only the mandatory datasets ensures a passing grade. To achieve a high score, you are expected to report experimental results on additional data.
- **Citations:** Ensure all related works and methods are correctly cited.

3 Guide to Start

To help you get started efficiently, we recommend the following workflow:

1. **Step 1: Literature Review.** Carefully read the papers listed in the **Recommended Reading List** (Section 4). Understanding the core concepts of 3DGS, Sparse Reconstruction, and Generative Reconstruction is crucial before coding.

2. **Step 2: Code Familiarization.** Clone the official repositories of the baseline methods (e.g., 3DGS). Run their provided demos to ensure your environment is set up and you understand the input/output formats.
3. **Step 3: Experimentation.** Run the methods on the **Mandatory Datasets** (see Section 6). Adjust code as required by the roadmap. Save your visualization results.
4. **Step 4: Writing & Submission.** Organize your results into tables and figures. Write your report using the CVPR template, citing **ALL** the references. Finally, submit the PDF report to Canvas.

4 Recommended Reading List

- **3DGS Reconstruction:** 3DGS [9], Scaffold-GS [14], Wavelet-GS [25]
- **3D Foundation Models:** VGGT [17], VGGT-Ω [18], DA3 [11], LingBot-Map [4], HorizonStream [5].
- **Unposed Sparse Reconstruction:** i3dgs [15], HI-SLAM2 [23], S3PO-GS [6].
- **Generative Reconstruction:** BRPO [24], Difix3D+ [20], GeoQuery [3], GenFusion [21], GSFixer [22].

5 Implementation Roadmap

5.1 Part 1: High-Fidelity Reconstruction (Initialization Analysis)

Motivation & Goal: Modern 3D Foundation Models are changing the traditional reconstruction pipeline from geometry-driven optimization toward data-driven feed-forward prediction. Compare classical SfM, a modern foundation model, and its further scaled version to understand this paradigm shift and the impact of model/data scaling.

Scenario: Given a video of an object or scene captured with dense views (sufficient data).

Technical Pipeline:

1. **Pose & Geometry Initialization:**
   - *Plan A (Traditional):* Run **COLMAP** [1] to estimate camera poses and a sparse point cloud.
   - *Plan B (Foundation Model):* Use **VGGT** [17] to predict camera poses and scene geometry.
   - *Plan C (Scaled Foundation Model):* Use **VGGT-Ω** [18] to predict camera poses and scene geometry.
2. **3DGS Optimization:**
   - Convert the estimated cameras/geometry to the required format and train the scene using the standard **3DGS** [9] pipeline.
   - **Key Analysis:** Compare initialization runtime, pose accuracy, 3DGS convergence speed, and final rendering quality across COLMAP, VGGT, and VGGT-Ω.
   - *Advanced Exploration (Optional):* Interested students may explore other recent 3D Foundation Models such as: **DA3** [11], **Pi3-X** [19], or streaming feed-forward models such as **LingBot-Map** [4] and **HorizonStream** [5].

Output: Compare three plans in runtime, pose accuracy, convergence, and novel-view rendering quality. Discuss how data-driven 3D reconstruction differs from classical multi-view geometry and what further scaling brings.

5.2 Part 2: Unposed Sparse Reconstruction

Motivation & Goal: Dense observations make reconstruction relatively well-constrained. This part studies the more challenging setting where only sparse views are available and camera poses are unknown.

Requirement: Choose **ONE** of the following options (Option A or Option B).

Scenario: Sub-sample the data from Part 1 to create extremely sparse frames (see Section 6), with **NO** camera poses provided.

Option A: Unposed Sparse Reconstruction (Recommended: i3dgs [15])

- **Full-View Reconstruction:** First run i3dgs on the complete input sequence to obtain a well-observed reconstruction.
- **Sparse-View Reconstruction:** Run the same pipeline again using the sparse sequence defined in Section 6, without providing camera poses.
- **Key Analysis:** Compare full-view and sparse-view reconstruction in pose accuracy, rendering quality, geometry completeness, runtime, and failure cases.
- *Challenge:* Maintaining reliable pose estimation and global consistency as view overlap becomes increasingly sparse.

Option B: Monocular 3DGS-SLAM (Recommended: HI-SLAM2 [23] or S3PO-GS [6])

- **System Loop:** Use a monocular 3DGS-SLAM system to jointly estimate camera motion and incrementally reconstruct a Gaussian map.
- **Key Components:** Focus on the interaction between **Tracking** (camera pose estimation) and **Mapping** (Gaussian scene update), together with global consistency and scale control.
- *Tip:* HI-SLAM2 requires sufficiently overlapping frames for initialization. For extremely sparse settings, you may use a short dense sequence at the beginning for initialization and apply sparse sampling to the remaining frames.
- *Challenge:* Preventing tracking failure, scale drift, and map degradation under sparse or long-sequence inputs.

Output: A 3DGS model reconstructed without COLMAP-provided camera poses, together with evaluation of pose estimation and rendering quality. For both options, explicitly compare the full-view and sparse-view settings.

5.3 Part 3: Closed-Loop Generative Reconstruction (Research)

Motivation & Goal: Sparse observations may not provide enough information for complete reconstruction. This part explores how generative priors can provide missing visual evidence and iteratively improve the reconstructed scene.

Scenario: Build upon the reconstruction framework used in Part 2, or another suitable reconstruction pipeline of your choice.

Technical Pipeline:

1. **View Selection:** Select target views where additional observations are needed. A simple default is to interpolate a viewpoint between two sparse input views.
2. **Generative Enhancement:** Choose **ONE** of the following routes:
   - *Route A – Image Diffusion Restoration:* Render the selected novel view and restore it using **GeoQuery** [3]. You may follow the reconstruction–generation pipeline of **BRPO** [24].
   - *Route B – Video Diffusion Prior:* Use **GSFixer** [22] or other models to restore or generate additional views along a selected trajectory. You may follow the reconstruction–generation loop of **GenFusion** [21].
   - Implement another reasonable generative reconstruction strategy based on related work or your own design.
3. **Closed-Loop Reconstruction:** Add the generated/restored views back into the reconstruction pipeline, update the scene, and repeat the view-selection and generation process. Complete at least **two generation–reconstruction rounds**.
4. **Consistency-Aware Generative Fusion (Advanced):** Generated views may be visually plausible but geometrically inconsistent. Design a strategy to detect, down-weight, reject, or correct inconsistent content, e.g., through confidence weighting, geometric alignment, or non-rigid deformation. Refer to **BRPO** [24] and **Video_to_world** [8].
5. **Efficient Video Diffusion (Advanced for Route B):** Reduce video-generation latency using training-free acceleration methods such as **TeaCache** [13] or **SenCache** [7].
6. **Active / Agentic View Planning (Advanced):** Replace fixed view interpolation with an active or agent-based strategy that selects where to generate next based on the current reconstruction.

Output: Compare the initial sparse reconstruction with the results after each generation–reconstruction round. Visualize the selected views, generated/restored observations, and the corresponding reconstruction improvements, and analyze success and failure cases.

6 Dataset & Evaluation

6.1 Datasets

- **Mandatory Data:** You can download from Baidu or Google.
  - `Waymo-405841` (Outdoor) – P2 Sparsity: 1/10 frames (follow BRPO)
  - `Dl3dv-2` (Outdoor) – P2 Sparsity: 1/30 frames (follow BRPO)
  - `Re10k-1` (Indoor) – P2 Sparsity: 1/30 frames
- **Optional Data:** Full Waymo [16]/Dl3dv [12] datasets mentioned in the S3PO-GS codebase, or Mip-NeRF 360 [2] / Tankand-Temple [10] datasets used in classic 3DGS.

6.2 Metrics

- **Rendering Quality:** PSNR, SSIM, LPIPS (calculated on the Test Set).
- **Pose Error (for Part 2):** ATE (Absolute Trajectory Error) RMSE — Compare the estimated poses estimated against the Ground Truth poses.

- **Runtime:** Compare the time required for pose/geometry initialization and the overall reconstruction process.

6.3 Qualitative Results

You must visualize and compare:

1. **Part 1:** Reconstruction quality and convergence among [COLMAP + 3DGS], [VGGT + 3DGS], and [VGGT-Ω + 3DGS].
2. **Part 2:** Reconstruction results from the Unposed Sparse Reconstruction Pipeline (Option A) or Monocular 3DGS-SLAM (Option B), including a Full Views vs. Sparse Views comparison.
3. **Part 3:** Reconstruction effects [With vs. Without] Generative View Completion.

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